3/23 CONSTRUCTING TOPOLOGIES FROM EXISTING ONES

Example: Let $\{\mathcal{T}_\alpha\}_{\alpha\in A}$ be a collection of topologies on X. Let X be finite. Then $\bigcap_{\alpha\in A}\mathcal{T}_\alpha$ is a bad idea for a topology on X.

Subspace Topology

Def Let $(X,\mathcal{T})$ be a topological space and $A\subseteq X$. Let $i_A : A \longrightarrow X$ be the standard inclusion ($i_A(a) = a$ for all $a\in A$). The *subspace topology* on A is $i_A^{-1}(\mathcal{T})$.

Rmk The way $\mathcal{B}\subseteq A$ is "open in A" if $\mathcal{B}$ is open in the subspace topology. Call "closed in A" if $A\setminus\mathcal{B}$ is open in the subspace topology.

Rmk I.e. the sliding down $\mathcal{B}\subseteq A$ is open in A iff $\mathcal{B} = A\cap U$ for some open set $U\subseteq X$.

Prop $\mathcal{B}$ is open in A. Now
Let $\mathcal{B}\in i_A^{-1}(\mathcal{T})$ for some open set $U\in\mathcal{T}$.
Hence, $i_A^{-1}(U) = U\cap A$ $\quad \forall U\in\mathcal{T}$.

Rmk If A is open, then $\mathcal{B}\subseteq A$ is open in A iff $\mathcal{B}$ is open in X.

Prop Given $A\subseteq X$.
$\Leftarrow$ If $\mathcal{B}\subseteq A$ open in X, then $\mathcal{B}\subseteq A\cap U$ for some U open in X.
Hence, $A\cup$ one has open in X, $\mathcal{B}\subseteq A\cap U = \mathcal{B}$ open in X.

Rmk The set A is left open and closed in A.

Eg I. $[0,1]$ for $A = [a,b]$,
The open sets of A are unions of $\mathcal{B}$ of the form
(1) open intervals (c,d) where $a\le c\le d\le b$

(2) intervals $[a,c)$ where $a<c\le b$

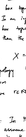
(3) intervals $(c,b]$ where $a<c\le b$

Defining Metric Topology on $\mathbb{R}^2$

Def Define the following order on $\mathbb{R}\times\mathbb{R}$ (where $\mathcal{B}$ denotes the set with an $\times$) $\mathcal{B}\subseteq\mathcal{B}$ if either $x<x_0$ or $x=x_0$ and $y\le y_0$

Let $B = \{(x_0,x_0) : x_0<x_0\}$ the following order topology on $\mathbb{R}^2$ is the topology generated by B.

Example B is closed in this order topology.



Product and Box Topology

Def Let $(X,\mathcal{T})$ and $(Y,\mathcal{T}')$ be topological spaces. Then the "product topology" $\mathcal{T}$ on $X\times Y$ is the smallest topology such that $\pi_X^{-1}(\mathcal{T})$ and $\pi_Y^{-1}(\mathcal{T}')$, where π_X and π_Y are the standard projection maps

$$\pi_X : X\times Y \longrightarrow X \qquad \pi_Y : X\times Y\longrightarrow Y$$
$$\pi_X(x,y) = x \qquad \pi_Y(x,y) = y$$

Rmk The sets of the form $U\times V$ where $U\in\pi_X^{-1}(\mathcal{T})$, $V\in\pi_Y^{-1}(\mathcal{T}')$ form a basis for $\mathcal{T}$.

Proof Let $\mathcal{B} = \{U\times V : U\in\mathcal{T}_X, V\in\mathcal{T}_Y\}$.
Claim $\mathcal{B}$ is a basis for a topology on $X\times Y$.
Let $\mathcal{T}'$ be the topology generated by $\mathcal{B}$.

Note that
$\pi_X^{-1}(U) = U\times Y$ $\quad \forall U\subseteq X$.
$\pi_Y^{-1}(V) = X\times V$ $\quad \forall V\subseteq Y$.
Then $\pi_X^{-1}(\mathcal{T}_X) = \{U\times Y : U\subseteq X\}$
$\pi_Y^{-1}(\mathcal{T}_Y) = \{X\times V : V\subseteq Y\}$.

So see that $\mathcal{T}\subseteq\mathcal{T}'$.
Note that $\pi_X^{-1}(\mathcal{T}_X)$ and $\pi_Y^{-1}(\mathcal{T}_Y)$ are made up of elements of $\mathcal{B}$. Thus $\mathcal{T}$, the smallest topology containing them, is contained in the topology generated by $\mathcal{B}$.

Hence $\mathcal{T}\subseteq\mathcal{T}'$.
Note that $\forall U\in\mathcal{T}_X, V\in\mathcal{T}_Y$,
$$U\times V = (U\times Y)\cap(X\times V) \in \mathcal{T}.$$
$$\pi_X^{-1}(\mathcal{T}_X) \qquad \pi_Y^{-1}(\mathcal{T}_Y)$$
$\therefore$ $\mathcal{B}\subseteq\mathcal{T}$.
$\Rightarrow \mathcal{T}'\subseteq\mathcal{T}$

Example: Consider the product topology $\mathcal{T}$ on $\mathbb{R}\times\mathbb{R}$ generated by two copies of the standard topology on $\mathbb{R}$.
At least for $\mathcal{T}$ is given by sets of the form $U\times V$ where U,V are open $\subseteq\mathbb{R}$. However, U,V are disjoint unions of open intervals.
Thus $U\times V$ is a disjoint union of open rectangles $(a,b)\times(c,d)$. Thus, $\mathcal{B}$ is the standard topology on $\mathbb{R}^2$.

Def Let $\{X_\alpha\}_{\alpha\in A}$ be a collection of sets. Let $\mathcal{B}$ be the set of all $U = \prod_{\alpha\in A} U_\alpha$ where $U_\alpha\subseteq X_\alpha$ for all $\alpha\in A$.
Let $\pi_\alpha : X\longrightarrow X_\alpha$ be the projection $\pi_\alpha(x) = x_\alpha$.
Eg: When $A = \mathbb{N}$, the elements of $\mathcal{B}$ are sequences $x = (x_1, x_2, \ldots)$ and $\pi_i(x) = x_i$ for $i\in\mathbb{N}$.

Def Let $\{(X_\alpha,\mathcal{T}_\alpha)\}_{\alpha\in A}$ be a collection of topological spaces.
The "product topology" on X is the smallest topology containing $\pi_\alpha^{-1}(\mathcal{T}_\alpha)$ $\forall\alpha\in A$.

Prop The product topology $\mathcal{T}$ on X is generated by sets of the form $U = \prod_{\alpha\in A} U_\alpha$ where
(1) $U_\alpha\in\mathcal{T}_\alpha$ $\forall\alpha\in A$
(2) $U_\alpha = X_\alpha$ for all but finitely many $\alpha\in A$.

Proof **Claim:** These sets form a basis for a topology $\mathcal{T}'$.
To see this $\mathcal{T}'$,
we all show each $\pi_\alpha^{-1}(\mathcal{T}_\alpha)\subseteq\mathcal{T}'$.
For each $U_\alpha\subseteq X_\alpha$ given
$$\pi_\alpha^{-1}(U_\alpha) = \prod_{\beta\in A} U_\beta \quad \text{where } U_\beta = X_\beta \text{ if } \beta\ne\alpha,\ U_\alpha = U_\alpha$$
Then $\pi_\alpha^{-1}(U_\alpha)\subseteq\mathcal{T}'$.
So see $\mathcal{T}'\subseteq\mathcal{T}$, let $U = \prod_{\alpha\in A} U_\alpha$ where $U_\alpha\subseteq X_\alpha$ for all $\alpha\in\ldots\ldots$.
So see $U = \pi_1^{-1}(U_1)\cap\pi_2^{-1}(U_2)\cap\ldots\cap\pi_n^{-1}(U_n)$.
$\in\mathcal{T}$.

Def The "box topology" on X is the topology generated by sets of the form $\prod_{\alpha\in A} U_\alpha$ where $U_\alpha\in\mathcal{T}_\alpha$ $\forall\alpha\in A$.

Rmk In finite product, the product and box topologies are the same.
Eg: In infinite product (eg $\mathbb{R}^{\mathbb{N}}$), the box topology is strictly finer (smaller) than the product topology.

Eg I. $X = \mathbb{R}^{\mathbb{N}}$: $\prod_{\mathbb{N}}\mathbb{R}$ the product topology is generated by sets of the form $U = U_1\times\ldots\times U_n\times\mathbb{R}\times\mathbb{R}\times\ldots$ where $U_1,\ldots,U_n$ are open intervals in $\mathbb{R}$.

Remark: The box construction will actually exercise that the product topology is the natural topology on a product space, under categorical observations.

Prop The product topology on X is the smallest topology for which all the projections π_α are continuous.

Proof $\pi_\alpha^{-1}(\mathcal{T}_\alpha)$ is the smallest topology on X for which π_α is continuous and $\mathcal{T}_\alpha$ is continuous for a topology $\mathcal{T}'$ (on $\mathcal{T}'$) contains all the $\pi_\alpha^{-1}(\mathcal{T}_\alpha)$ (so $\mathcal{T}'$ contains the product topology $\mathcal{T}$)

Exercise: Let $(X,\mathcal{T})$ be a topological space. $f : X\longrightarrow Y$ is a function. Then $\{V\subseteq Y : f^{-1}(V)\in\mathcal{T}\}$ is a topology $= \mathcal{T}'$.
Then is the largest topology $= \mathcal{T}$ for which f is continuous.

Remark: This is not $\{f(V)\subseteq Y : V\in\mathcal{T}\}$.
It is not hard for topology $\{f_*\mathcal{T}\}$ and not $f(\mathcal{T})$ as for surjective.

Quotient Topology

Def Let X and Y be topological spaces and $p : X\to Y$ be a surjective function.
The "quotient map" p if $V\subseteq Y$, V is open in Y iff $p^{-1}(V)$ is open in X.
(In particular, p is continuous).

Example If X_1 and X_2 are the topological spaces, this is equivalent to the condition that $X_2 = X_1/X_1$.

Def Let A be a set and X a topological space, and $p : X\to A$ be surjective function. The quotient topology on A is the largest topology for which p is a quotient map.

Rmk $B\subseteq A$ open iff $p^{-1}(B)$ is open in X.
a quotient topology.

Eg I. Let $(X,\mathcal{T})$ be a topological space, X^* be a partition of X into disjoint subsets. Define X^* to X.
Consider the function $p : X\longrightarrow X^*$ s.t. $p(x)\in X^*$ is the unique partition element containing x.
The quotient topology on X^* is $p_*\mathcal{T}$.
$U\subseteq X^*$ is open iff $p^{-1}(U)$ is open.
The subspace topology on $[0,1]$.

To illustrate, consider the partition X^* of $X = [0,1]$ given by $X^* = \{\{x\} : x\in[0,1]\}$ $\cup$ $\{\{x,1-x\} : x\in[0,\frac{1}{2}]\}$ (I.e. the partition is identified).
Then the quotient topology on X^* makes it "look" like a circle, in a sense that we will make precise later on.
$p(x) = \begin{cases}\{x\} & x\ne 0,1\\ \{x,1\} & x=0,1\end{cases}$

Metric Topology - revisited

Prop Let X be a set and $d : X\times X\longrightarrow[0,\infty)$ be a metric.
Let $\mathcal{T}$ be the topology generated by $\mathcal{B}$.
Let $\mathcal{T}'$ be the topology generated by the function $d(x,y) : X\longrightarrow\mathbb{R}$.
The metric topology on X is the coarsest topology containing $\{d_x^{-1}(\mathcal{T})\}_{x\in X}$ (where $\mathcal{T}$ is the subspace topology on $[0,\infty)$).

Remark: The metric topology is the coarsest topology on X for which each d_x is continuous.

Example Let (X,d) be a metric space, and consider the product topology on $X\times X$, namely the metric topology on $\mathcal{T}$.
Then $d : X\times X\longrightarrow[0,\infty)$ is continuous.

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SEQUENCES

Def: Let X be a set. A "sequence" in X is a function $x: \mathbb{N} \longrightarrow X$ i.e., an infinite collection of points $x_1, x_2, x_3, \ldots$ $(x_n = x(n))$.

Def: Let $(X, \mathcal{T})$ be a topological space and let $x \in X$. We say that the sequence $(x_n)_{n \in \mathbb{N}}$ "converges to"/"tends to" x and write $x_n \longrightarrow x$, if for every neighborhood U of x, $\exists N \in \mathbb{N}$ s.t. $x_n \in U \ \forall n \geq N$.

Eg: 1. In $\mathbb{R}$ with the std. topology,

$$1/n \longrightarrow 0,$$

$$\frac{1}{n^{1+\varepsilon}} \longrightarrow 0 \ \forall \varepsilon > 0$$

$$\left(1 + \frac{1}{n}\right)^n \longrightarrow e$$

2. Sequences in general spaces may not converge at all, or may converge to more than one point.

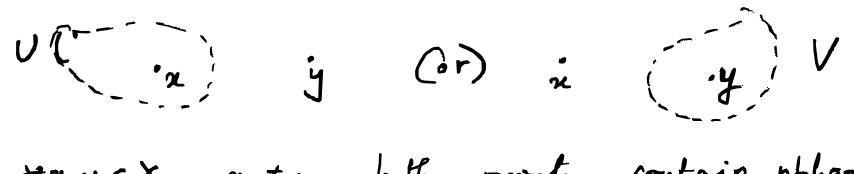
* In $\mathbb{R}$ with the std topology, the sequence $x_n = n$ does not converge.
* In the indiscrete topology $\mathcal{T} = \{\phi, X\}$ on any set X, every sequence converges to every point.
* Line with doubled origin: Consider the non Hausdorff space $\mathbb{R} \setminus \{0\} \cup \{0_+, 0_-\}$ introduced in a previous lecture. Here, $1/n \longrightarrow 0_+, 0_-$.

Separation Axioms

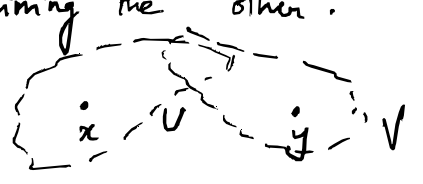
These are properties that measure how "separated" the closed sets of a topological space are.

A topological space $(X, \mathcal{T})$ is said to be:

* T_0, if $\forall x, y \in X$, $x \neq y$, $\exists$ an open nbhood of at least one of the points that does not contain the other.



* T_1, if $\forall x, y \in X$, $x \neq y$, both points contain nbhoods not containing the other.



* T_2, if $\forall x, y \in X$, $x \neq y$, both points have nbhoods that are disjoint.

Eg:

1. A topological space that is T_0, but not T_1:
$\mathbb{R}$, with a topology where the open sets are of the form (a, ∞)
Given $x \neq y$ (WLOG $x < y$).
$\left(\frac{x+y}{2}, \infty\right)$ contains y but not x.
However it is not possible to find an open set containing x but not y.

2. Space that is T_1, but not T_2
Cofinite topology on $\mathbb{R}$:
No two open sets can be disjoint so this space is not T_2.
Given $x \neq y$ in $\mathbb{R}$, $\mathbb{R} \setminus \{x\}$ is an open nbhood of y that does not contain x, and $\mathbb{R} \setminus \{y\}$ is an open nbhood of x not containing y.

Remark: $T_2 \Rightarrow T_0 \Rightarrow T_1$. [There are T_n spaces for $n \geq 3$, but we will discuss these later]

Sequences and Limit points

Fix a T_1-space $(X, \mathcal{T})$ and let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X, and suppose $x \in X$.

Prop: For any $A \subseteq X$, $x \in L(A) \Leftrightarrow$ every nbhood of x contains infinitely many points of A.

Proof:

Prop: Let $A = \{x_1, x_2, x_3, \ldots\}$.
Then $x \in L(A) \Leftrightarrow \exists$ a subsequence $x_{n_k} \longrightarrow x$.

Proof:

Prop: If X is T_2 (Hausdorff), any sequence of points converges to at most one point in X.

Proof:

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CONTINUOUS FUNCTIONS, REVISITED

Prop: For metric spaces (X,d_X) and (Y,d_Y), the ε-δ definition of a continuous function and the topological definition coincide.

Proof:

Standard Examples

Let X, Y and Z be topological spaces.

1. Constant function: Let $f: X \longrightarrow Y$ be a function s.t. for some $y_0 \in Y$, $f(x) = y_0 \; \forall x \in X$. Then f is continuous, $\because$ the inverse image of any set is either $\emptyset$ or X.

2. Restricting Domain: If $f: X \to Y$ is continuous and $A \subseteq X$, then in the subspace topology, the restricted function $f|_A : A \to Y$ is continuous.

   Proof: Let $g = f|_A$. $\forall V \subseteq Y$ open, $g^{-1}(V) = f^{-1}(V) \cap A$, which is open in A.

3. Restricting / Expanding Range: Let $f: X \longrightarrow Y$ be continuous (i) Given a subset Z of Y s.t. $f(X) \subseteq Z$, the function $g: X \to Z$ obtained by restricting the range of f is continuous.

   (ii) If $Y \subseteq Z$, then $h: X \to Z$ obtained by expanding the range of f is continuous.

   Proof: (i) Given $V \subseteq Z$ open in Z, $V = V' \cap Z$ for some open set $V' \subseteq Y$. Then note that $g^{-1}(V) = f^{-1}(V')$, which is an open subset of X.

   (ii) Left as an exercise.

4. "Local" formulation: The map $f: X \longrightarrow Y$ is continuous iff $\exists$ a collection $\{U_\alpha\}_{\alpha \in A}$ of open sets in X s.t. $\bigcup_{\alpha \in A} U_\alpha = X$ and $f|_{U_\alpha}$ is continuous for each α.

   Proof: Assuming such a collection exists, $\forall V \subseteq Y$ open,

   $$f^{-1}(V) = \bigcup_{\alpha \in A} (f^{-1}(V) \cap U_\alpha) = \bigcup_{\alpha \in A} (f|_{U_\alpha})^{-1}(V)$$

   Note: each set $(f|_{U_\alpha})^{-1}(V)$ is open in U_α $\because f|_{U_\alpha}$ is cts. Since U_α is open in X, each of these sets is also open in X, and thus their union, $f^{-1}(V)$, is open in X.

5. Composition: Let $f: X \longrightarrow Y$ and $g: Y \longrightarrow Z$ be continuous. Then $g \circ f: X \to Z$ is continuous.

   Proof: $\forall V \subseteq Z$ open, $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$. $\because g^{-1}(V) \subseteq Y$ is open, $f^{-1}(g^{-1}(V)) \subseteq X$ is open.

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Homeomorphisms

Def: Let $f: X \longrightarrow Y$ be a bijection. The map f is said to be a "homeomorphism" if f and its inverse $f^{-1}: Y \to X$ are both continuous.

Eg: 1. Non-constant linear functions on $\mathbb{R}$: Consider any function $f: \mathbb{R} \longrightarrow \mathbb{R}$ given by $f(x) = mx + b$ where $m \neq 0$. Then f is a homeomorphism.

2. Consider the function

$$f: (-1,1) \longrightarrow \mathbb{R}, \quad x \longmapsto \frac{x}{1-x^2}$$

This is a homeomorphism.

Proof:

3. Let $B_n = \{x \in \mathbb{R}^n : d(x,0) < 1\}$. Then the function $F: B_n \longrightarrow \mathbb{R}^n$, $x \longmapsto \frac{x}{1-(d(x,0))^2}$ is a homeomorphism.

Proof:

Exercise: Show that any two open intervals in $\mathbb{R}$ are homeomorphic.

Non-example: 1. Consider the identity function $f: \mathbb{R} \longrightarrow \mathbb{R}_\ell$, $[f(x) = x \;\; \forall x \in \mathbb{R}]$. Then f is not continuous: $[a,b)$ is open in $\mathbb{R}_\ell$, but $f^{-1}([a,b)) = [a,b)$ is not open in $\mathbb{R}$.

Exercise: For this f, show that $f^{-1}: \mathbb{R}_\ell \longrightarrow \mathbb{R}$ is continuous.

2. Let $S^1 = \{(x,y) \in \mathbb{R}^2 : x^2+y^2=1\}$ denote the unit circle. Consider $f: [0,1) \longrightarrow S^1$ given by $f(t) = (\cos(2\pi t), \sin(2\pi t))$. Then f is continuous, but not a homeomorphism and bijective.

Proof:

Imbeddings

Def: Let $f: X \longrightarrow Y$ be an injective continuous map. Suppose $Z = f(X)$ and consider $f': X \to Z$ obtained by restricting the range of f.

Note: f' is a bijection.

If f' is a homeomorphism, we say f is a "(topological) imbedding" of X in Y.

Eg: 1. Let X, Y be topological spaces, and suppose $y_0 \in Y$. The map $f: X \longrightarrow X \times Y$ given by $f(x) = (x, y_0)$ is a topological imbedding.

Proof:

2. Let X be a topological space and A be a subspace of X. The inclusion $i_A : A \hookrightarrow X$ is an imbedding.

Proof: Exercise.

Non-eg: 1. Consider $f: [0,1) \longrightarrow S^1$, $x \longmapsto (\cos(2\pi x), \sin(2\pi x))$ as above, the function $f: [0,1) \longrightarrow \mathbb{R}^2$ obtained by expanding the range of f is not an imbedding of $[0,1)$ in $\mathbb{R}^2$.

Coordinate Functions

Theorem: Let $f: A \longrightarrow X \times Y$ be of the form $f(x) = (f_1(x), f_2(x))$ for some functions $f_1: A \to X$, $f_2: A \to Y$. [f_1, f_2 are called the coordinate functions of f.] Then f is continuous $\Leftrightarrow$ f_1 and f_2 are continuous.

Proof:

Uniform Limit Theorem